Rain Zhang

Full-Stack Developer · Simon Fraser University

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EXPERIENCE

Software and IT Systems Specialist

MNT Realty · Vancouver, BC, Canada — Aug 2026 – present

- Responsible for the company's internal systems, from requirements and design to hosting, deployment and maintenance.
- Planned and rebuilt the MNT Realty platform from scratch as one application for three sites: a public website, an owner portal for residents, and an admin console for staff.
- Designed and built MNT Control Center, an internal platform in Next.js, Node.js and PostgreSQL that brings the office's separate tools into one place.
- Set up organization sign-in and role-based access with Microsoft Entra ID, the Graph API and OAuth 2.0.
- Automated recurring, time-consuming work: sorting incoming email, looking up owner information, and answering staff questions from company documents.

Full-Stack Engineer Intern

FEITIAN Technologies · International Dept. · Beijing — Sep – Dec 2025

- Planned, built and ran three production systems for FEITIAN's post-quantum FIDO2 work: a public WebAuthn developer platform, a software security key in Rust, and a customer demo site.
- Wrote a CTAP2.1 authenticator in Rust that Linux presents as a virtual USB security key, letting browsers test ML-DSA credentials ahead of the hardware.
- Deployed with Docker on Linux servers and Google Cloud Run; GitHub Actions ran tests, builds and a daily FIDO metadata refresh.
- Worked with hardware and security engineers so the tools matched real devices; still maintaining all three after the internship ended.

PROJECTS

MNT Realty Platform

Next.js · TypeScript · Node.js · PostgreSQL · Vercel · mntrealty.vercel.app — Aug 2026 – present

- One application serving three sites: the public website, an owner portal for residents, and an admin console for staff, each on its own subdomain.
- Routing by hostname gives each site its own layout and session cookie, and all reads and writes go through one data store interface.
- Navigation, page guards and file routes all read one access table, so users cannot see what other roles have access to.

WebAuthn Developer Platform

Python · Flask · Docker · liboqs · webauthndev.com · repository — Sep 2025 – present

- Public tool for FIDO2 developers, built because the team's tools could not show post-quantum credentials: register and sign in with real or virtual authenticators, decode responses, and search the FIDO Alliance metadata service.
- Added ML-DSA-44, -65 and -87 to a fork of Yubico's python-fido2 through liboqs, while keeping the classical algorithms working.
- About 120 server test files plus front-end and post-quantum suites run in CI; a daily GitHub Action re-verifies the metadata snapshot.

SKILLS

Languages
Python, TypeScript, JavaScript, Rust, C, C++, Java

Web
React, Next.js, Node.js, Flask, Tailwind CSS, HTML, CSS, REST APIs

Data and cloud
PostgreSQL, Docker, Google Cloud, Vercel, GitHub Actions, Git, Linux

Security
WebAuthn, FIDO2, CTAP2.1, OAuth 2.0, Microsoft Entra ID, post-quantum cryptography (ML-DSA, liboqs)

Testing
pytest, Vitest, Playwright, CI on GitHub Actions

EDUCATION

Simon Fraser University

Bachelor of Science, Computer Science
Vancouver, BC, Canada

Sep 2023 – Dec 2027 · CGPA 3.44 / 4.33
Dean's Honour Roll, Fall 2024 and Summer 2025